

MTH00050: Mathematical Combinatorics

Lecture 3: Inclusion-Exclusion Principle

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(Tentative) Schedule 2026

No.	Date	Location	Topic
1	13/01 16/01	C24 C24/E302	Introduction, Sets, Proofs
2	20/01 23/01	C24 C24/E302	General counting methods
3	27/01 30/01	C24 C24/E302	Inclusion-exclusion principle
4	03/02 06/02	C24 C24/E302	Recurrence relations
5	03/03 06/03	C24 C24/E302	Generating functions
			Mid-term test

No.	Date	Location	Topic
6	10/03 13/03	C24 C24/E302	Graphs (Part I)
7	17/03 20/03	C24 C24/E302	Graphs (Part II)
8	24/03 27/03	C24 C24/E302	Path problems
9	31/03 03/04	C24 C24/E302	Tree problems
10	07/04 10/04	C24 C24/E302	Network flows (Part I)
11	14/04 17/04	C24 C24/E302	Network flows (Part II)
			Final exam

Course topics

0. Introduction
1. Sets
2. General counting methods for arrangements and selections
3. Inclusion-exclusion principle
4. Recurrence relations
5. Generating functions
6. Graphs
7. Path problems
8. Tree problems
9. Network flows

Outline

1. Introduction
2. Terminologies
3. Venn Diagram method
4. Inclusion-Exclusion principle

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Venn Diagram Method

Proposition 1. Given finite sets A, B and U such that $A, B \subset U$ (U is called the *universal set*). In particular, $\bar{A} = U \setminus A$ and $\bar{B} = U \setminus B$ are the *complements* of A and B with respect to U . Then, we have:

a) $|A \cup B| = |A| + |B| - |A \cap B|.$

b) From $\bar{A} \cap \bar{B} = \overline{A \cup B} = U \setminus (A \cup B)$, we obtain:

$$|\bar{A} \cap \bar{B}| = |U| - |A \cup B| = |U| - (|A| + |B|) + |A \cap B|. \quad (1)$$

Example 1 (1/2)

Example 1. In 2025, at FIT-HCMUS, we have five classes with 100 students, of which 50 students know English, 40 students know French, and 20 students know both English and French. How many students at the school do not know either English or French?

Example 1 (2/2)

Solution. Let U be the student set, A be the set of students know English, and B be the set of students know French, we need to calculate $|\bar{A} \cap \bar{B}|$, so we have: $|\bar{A} \cap \bar{B}| = |U| - (|A| + |B|) + |A \cap B| = 100 - (50 + 40) - 20 = 30$.

Example 2 (1/2)

Example 2. *How many permutations* of the digits 0, 1, 2, ..., 9 are there such that the first digit is *greater than* 1 and the last digit is *less than* 8?

Example 2 (2/2)

Solutions. Let U be the set of all permutations of $0, 1, 2, \dots, 9$; A be the set of permutations starting with 0 or 1; and B be the set of permutations ending with 8 or 9.

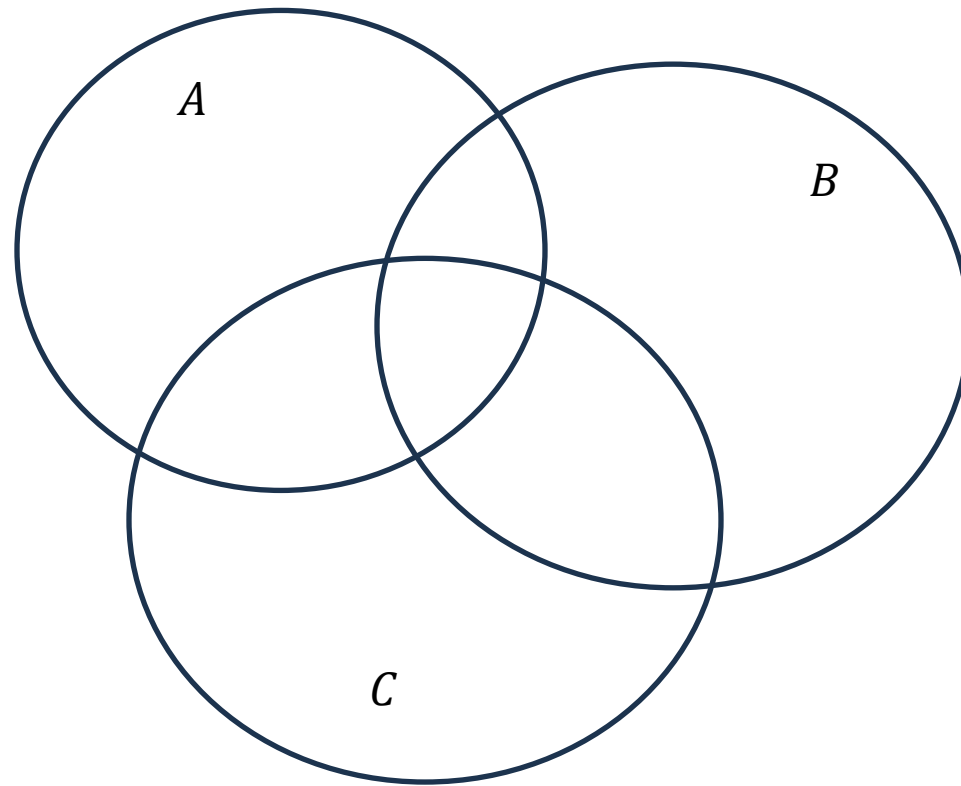
To solve this problem, we need to calculate $|\bar{A} \cap \bar{B}|$.

Apply the formula (1), we have:

$$\begin{aligned} |\bar{A} \cap \bar{B}| &= |U| - |A \cup B| = |U| - (|A| + |B|) + |A \cap B| \\ &= 10! - (2 \times 9!) - (2 \times 9!) + (2 \times 2 \times 8!) = 2.338.560 \end{aligned}$$

Venn Diagram Method

Question 1. How to extend the formula (1) for three sets A, B and C ?



$$|\bar{A} \cap \bar{B}| = |U| - |A \cup B| = |U| - (|A| + |B|) + |A \cap B|. \quad (1)$$

Venn Diagram Method

Question 1. How to extend the formula (1) for three sets A, B and C ?

Answer. We have

$$\begin{aligned} |\bar{A} \cap \bar{B} \cap \bar{C}| &= |U| - |A| - |B| - |C| \\ &\quad + |A \cap B| + |A \cap C| + |B \cap C| \\ &\quad - |A \cap B \cap C| \quad (2) \end{aligned}$$

Venn Diagram Method

Proposition 2. Given three finite sets A_1, A_2 and A_3 , we have:

$$\begin{aligned} & |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| \\ &= |U| - \sum_{i=1}^3 |A_i| + \sum_{1 \leq i \leq j \leq 3} |A_i \cap A_j| - |A_1 \cap A_2 \cap A_3| \quad (3) \end{aligned}$$

Example 3

Example 3. A school has *100 students*, of which *40* students study English, *40* students study French, *40* students study German. Each pair of languages is studied by *20* students, and *10* students study all three languages. *How many students do not study any of the three languages: English, French, and German?*

Venn Diagram Method

Remark 1. The number of positive integers $\leq n$ that are divisible by k is $[n/k]$.

Example 4

Example 4. How many positive integers less than 1000 such that divisible by 7?

Venn Diagram Method

Solutions. Apply the **Remark 1**, we have:

$$\left[\frac{999}{7} \right] = 142.$$

Example 5 (1/5)

Example 5. How many positive integers that less than or equal 1000 and co-prime with 70?

Venn Diagram Method

Example 6. How many positive integers that less than or equal 2000 and co-prime with 200?

Solution: $200 = 2^3 \times 5^2$

The number of positive integers requires is: $2000 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 800$

Example 7. How many positive integers that less than or equal 2000 and co-prime with 396?

Answer: 606.

Outline

1. Introduction
2. Terminologies
3. Venn Diagram method
4. Inclusion-Exclusion principle

Inclusion-Exclusion principle

In this section we will extend the formula in previous section to the case of n sets $A_1, A_2, \dots, A_n$. To simplify the notation, we write “ $\cap$ ” as multiplication. For example, $A_1 \cap A_2 \cap A_3$ will be written as $A_1 A_2 A_3$. Using this notation, we get the number of elements not belonging to all sets $A_1, A_2, \dots, A_n$ will be written as $|\overline{A_1} \times \overline{A_2} \dots \times \overline{A_n}|$.

Inclusion-Exclusion principle

Theorem 1. Let U be the universal set, and $A_1, A_2, \dots, A_n$ be n subsets of U . Set S_k be the sum of elements of all intersections of exactly k sets from $\{A_i\}_{i=1, \dots, n}$. In particular,

$$S_1 = \sum_{i=1}^n |A_i|, S_2 = \sum_{1 \leq i < j \leq n} |A_i A_j|, S_\ell = \sum_{1 \leq i_1 < \dots < i_\ell \leq n} |A_{i_1} \dots A_{i_\ell}|, S_n = |A_1 A_2 \dots A_n|.$$

Then,

$$|\overline{A_1} \overline{A_2} \dots \overline{A_n}| = |U| + \sum_{k=1}^n (-1)^k S_k.$$

Inclusion-Exclusion principle

Corollary 1. Given $A_1, A_2, \dots, A_n$ be n subsets of U . Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k-1} S_k$$

Inclusion-Exclusion principle

Corollary 1. Given $A_1, A_2, \dots, A_n$ be n subsets of U . Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k-1} S_k$$

Proof. By apply the Theorem 1, we have:

$$\begin{aligned} |\overline{A_1} \dots \overline{A_n}| &= |U| - S_1 + S_2 - \dots + (-1)^k S_k + \dots + (-1)^n S_n \\ &= |U| - (S_1 - S_2 + \dots + (-1)^k S_k + \dots + (-1)^{n-1} S_n). \end{aligned}$$



Example 8 (1/6)

Example 8. Find the number of non-negative integer solutions of this given equation:

$$x_1 + x_2 + x_3 + x_4 = 18 (*)$$

such as

$$x_i \leq 7, \forall i = 1, 2, 3, 4.$$

Example 9 (1/3)

Example 9. How many ways are there to draw 6 cards from a 52-card deck such that:

- a) all 4 suits (hearts, diamonds, clubs, and spades) are represented.
- b) at least one suit is not represented.

Example 10 (1/2)

Example 10. Find the number of non-negative integer solutions of this given equation:

$$x_1 + x_2 + x_3 + x_4 = 20$$

such as

$$x_1 \leq 5, x_2 \leq 7, x_3 \leq 10.$$

Example 11 (1/4)

Example 11. How many surjective functions are there from a set with 6 elements to a set with 3 elements?

Inclusion-Exclusion principle

Theorem 2. Given m and n be two positive integer such that $m \geq n$. Then, the number of surjective functions from the set of m elements to the set of n elements is:

$$n^m - \binom{n}{1} (n-1)^m + \binom{n}{2} (n-2)^m - \dots + (-1)^{n-1} \binom{n}{n-1} (1)^m$$

Example (1/3)

Example 12. How many ways are there to assign 5 different tasks to 4 people with the condition that each person has at least one task?

Inclusion-Exclusion principle

Theorem 3. Let U be the universal set, and $A_1, A_2, \dots, A_n$ be n subsets of U . Then, the number of elements belongs exactly m subsets, denote N_m , is calculated:

$$N_m = \sum_{i=0}^{n-m} (-1)^i \binom{m+i}{m} S_{m+i}$$

Inclusion-Exclusion principle

Theorem 3 (cont). Let N_m^* be the number of elements that belongs to at least m subsets. Then,

$$N_m^* = \sum_{i=0}^{m=n} (-1)^i \binom{m+i-1}{m-1} S_{m+i}$$

Example 13 (1/3)

Example 13. How many ternary sequences (consisting only of 0, 1, 2) of length 4

a) contain exactly 2 digits 1

b) contain at least 2 digits 1

Example 14 (1/5)

Example 14. There are 5 letters and 5 envelopes with pre-addressed labels. Each envelope is randomly filled with exactly one letter. What is the probability that **no letter is placed in the correct envelope**?

Generalize the problem with n (positive integer) letters and n envelopes.

Example 15 (1/3)

- There are 30 students in a dormitory.
- 15 students study painting, 8 students study biology, and 6 students study chemistry.
- It is known that 3 students take all three courses.
- Prove that at least 7 students do not take any course.

Q & A